Course Title & Course Code : Computer Methods in Power System Analysis (8EP02)

Class: Final year Electrical Semester: VIII

Name of the Course Teacher: Mr. R.K.Mankar

Title of the innovative practice: Experiment (simulation based) Development & Testing.

Students design new experiments, analyse results, and enhance laboratory skills.

Objectives of the practice:

i. The primary objective of this innovative teaching practice is to improve conceptual clarity and self-paced learning through the use of modern tool. (Simulation software)

ii. Students should compare results obtain from simulation with mathematical modelling of power system component (Transmission line)

The specific goals are:

i. To make complex topics in Computer Methods in Power System Analysis more accessible and engaging.

ii. To support diverse learning styles through simulation

iii. To provide students with a platform for revision and reinforcement outside of the classroom.

iv. Develop, simulate and test an accurate transmission line model to analyse overall system performance - without the need for physical prototyping.

Use of Appropriate Tool : Desktop Computer with MATLAB Simulation Software.

Creation of Topic through Simulation : Simulation were created covering determination of sending end voltage, power and voltage regulation of transmission line using MATLAB software.

Interactive Content: The simulation included modelling of Transmission line.

Follow-up Sessions: In-class discussions and Q&A sessions were conducted to clarify

doubts after session.

Improved Understanding: Students reported better conceptual understanding of difficult

topics as needed.

Flexible Learning: The simulation by using MATLAB software includes modelling of Transmission line and compared results obtained from simulation with Analytical results.

Extended Reach: Modelling of power system components by using simulation software can be carry out anytime and anywhere, even beyond college working hours.

POs Mapped: PO1, PO2, PO5, PO12

Reflective Critique:

Faculty observed increased student interaction and curiosity during lectures that followed the sessions. Students appreciated the convenience and clarity offered by the simulation . Feedback from students included suggestions for use of other simulation software (namely PSCAD, ETAP) for power system analysis.

Evidences of Success:

Most of the students performed modelling of various power system components through simulation software . It helps students to compare results obtained from simulation with Analytical (Mathematical) .

Peer Review & Critique Form Link: https://forms.gle/v9co2VzdwBAZYS86A

Numerical : A 220 kV, 3-phase transmission line is 40 km long. The resistance per phase is 0.15 Ω per km and the inductance per phase is 1.3263 mH per km. The shunt capacitance is negligible. Use the short line model to find the voltage and power at sending end and the voltage regulation when the line supplying three phase load of 381 MVA at 0.8 power factor lagging at 220 kV.

Case Study :

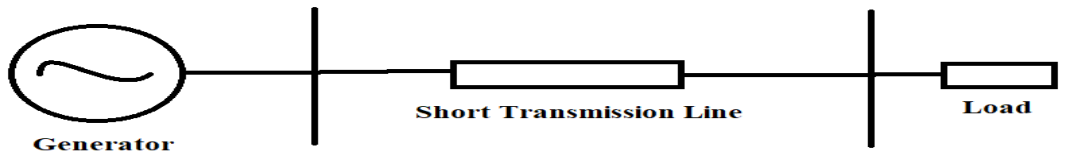

Fig. a):Single Line Diagram of Short Transmission Line Model

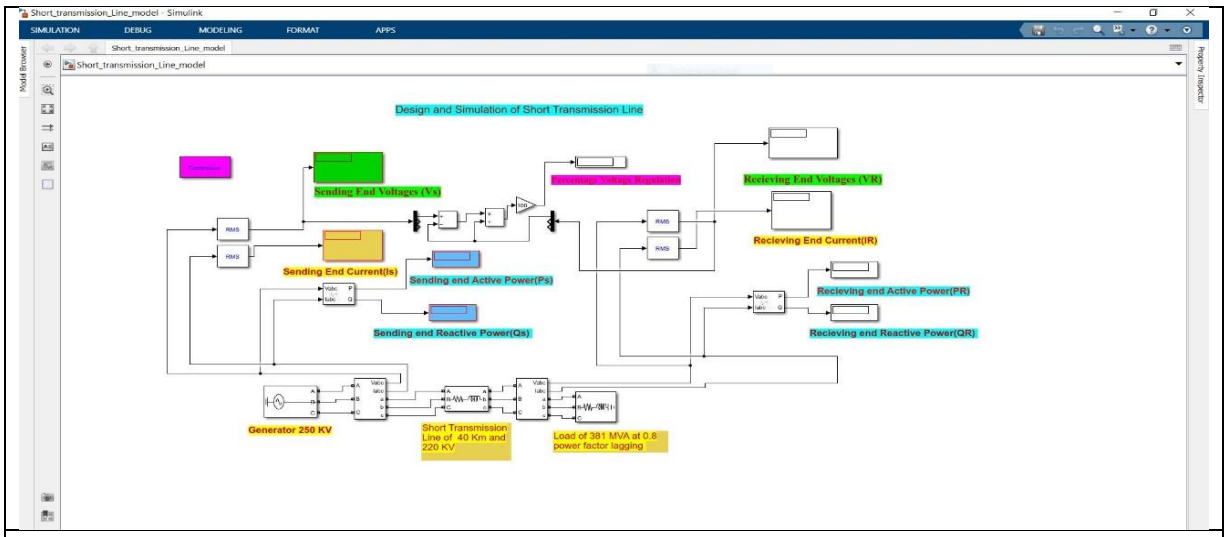

Fig b: Design and simulation of short Transmission line by using MATLAB software

Solution :

The series impedance per phase is

$$Z = (r + j\omega L) \times l = (0.15 + j2\pi \times 60 \times 1.3263 \times 10^{-3)}\ 40 = 6 + j\ 20\ \Omega$$

The receiving end voltage per phase is

$$V_R = \frac{220\angle 0^\circ}{\sqrt{3}} = 127\angle 0^\circ \text{ kV}$$

The apparent power is

$$S_{R(3\Phi)} = 381\angle \cos^{-1} 0.8 = 381\angle 36.87^\circ = 304.8 + j228.6 \text{ MVA}$$

The current per phase is given by

$$I_R = \frac{S^*_{(3\emptyset)}}{3V_R^*} = \frac{381\angle 36.87^\circ \times 10^3}{3 \times 127\angle 0^\circ} = 1000\angle -36.87^\circ \text{ A}$$

The sending end voltage is

$$V_S = V_R + ZI_R = 127\angle 0^\circ + (6 + j20)\,(1000\angle -36.87^\circ)(10^{-3})$$

$$= 144.33\angle 4.93^\circ \text{ kV}$$

The sending end line-to-line voltage magnitude is

$$|V_{S\,(L\text{-}L)}| = \sqrt{3}|V_S| = 250 \text{ kV}$$

The sending end power is

$$S_{S\,(3\Phi)} = 3V_S I_S^* = 3 \times 144.33\angle 4.93 \times 1000\angle 36.87^\circ \times 10^{-3}$$

$$= 322.8 \text{ MW} + j288.6 \text{ MVar}$$

$$= 433\angle 41.8^\circ \text{ MVA}$$

% Voltage regulation = $\frac{250-220}{220} \times 100 = 13.6\%$

System Parameters:

A) Generator Specification:

i) Sending end voltage (L-L): 250 kV (calculated)

ii) Sending end voltage per phase: $144.33\angle 4.93^\circ$ kV (calculated)

iii) Sending end per phase Current: $1000\angle -36.87^\circ$ A (calculated)

iv) System Frequency: 60Hz

B) Transmission Line Specification:

i) Line length: 40 km

ii) Line voltage: 220kV (L-L)

iii) Inductance: 0.053052 H

iv) Resistance: 6 Ω

v) Shunt Capacitance: 0 Farad

C) Load Specification:

i) Nominal phase to phase voltage:220kV

ii) Active Power (P): 304.8 MW

iii) Reactive Power (Q): 228.6 MVar

Procedure:

i) Open the MATLAB window by clicking the MATLAB icon and from the MATLAB window click on the Simulink toolbox to open the SIMULINK environment.

ii) Open the Library browser and add the appropriate component/blocks in the Simulink environment which are required for the system.

iii) Connect the component as per fig. b).

iv) Double click on each component and edit the system parameters.

v) Set 1 sec run time in Simulink window .

vi) Run the Simulation and verify the simulated results with the analytical results obtained.

Results:

Sending End							
Voltage (kV)		**Active Power (P_S) (MW)**		**Reactive Power (Q_S) (MVar)**		**% Voltage Regulation**	
Analytical	**Simulated**	**Analytical**	**Simulated**	**Analytical**	**Simulated**	**Analytical**	**Simulated**
144.33∠4.93°	144	322.8	322.8	288.6	288.5	13.6	13.5

References:

[1] Hadi Saadat, “Power System Analysis”, published by Tata McGraw Hill.

[2] www.mathsworks.com